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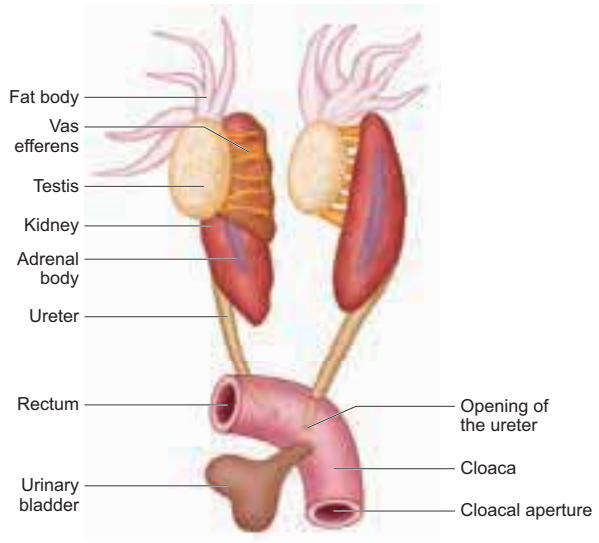


Figure 4.23 *Rana hexadactyla* - Male Urinogenital System

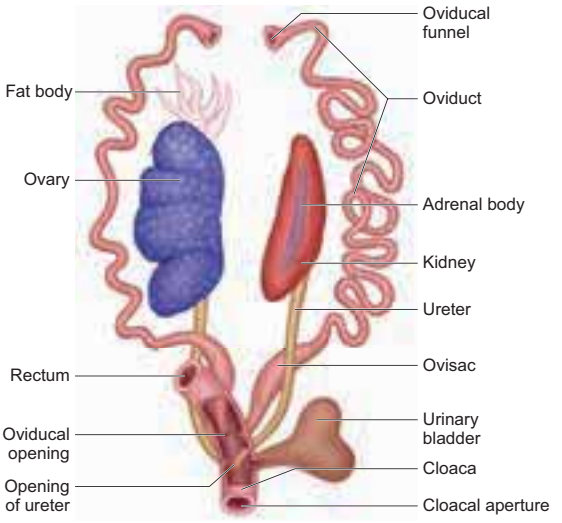


Figure 4.24 *Rana hexadactyla* - Female Urinogenital System

contain a small cavity called **olfactory ventricle**. The mid brain (Mesencephalon) includes two large, oval **optic lobes** and has cavities called **optic ventricles**. The hind brain (Rhombencephalon) consists of the **cerebellum** and **medulla oblongata**. Cerebellum is a narrow, thin transverse band followed by **medulla oblongata.** The medulla oblongata passes out through the **foramen magnum** and continues as **spinal cord,** which is enclosed in the vertebral column (Figure 4.22).

Excretory system

Elimination of **nitrogenous waste** and salt and water balance are performed by a well developed excretory system. It consists of a pair of kidneys, ureters, urinary bladder and cloaca. Kidneys are dark red, long, flat organs situated on either sides of the vertebral column in the body cavity. Kidneys are **Mesonephric.** Several nephrons are found in each kidney. They separate nitrogenous waste from the blood and excrete urea, so frogs are called **ureotelic** organisms. A pair of ureters emerges from the kidneys and opens into the cloaca. A thin walled unpaired **urinary bladder** is present ventral to the **rectum** and opens into the **cloaca.**

Reproductive system

The **male** frog has a pair of testes which are attached to the kidney and the dorsal body wall by folds of peritonium called mesorchium. Vasa efferentia arise from each **testis**. They enter the kidneys on both side and open into the bladder canal. Finally, it communicates with the urinogenital duct that comes out of kidneys and opens into the cloaca (Figure 4.23).

Female reproductive system (Figure 4.24) consists of paired **ovaries,** attached to the kidneys, and dorsal body wall by folds of peritoneum called mesovarium. There is a pair of coiled **oviducts** lying on the sides of the kidney. Each oviduct opens into the body-cavity at the anterior end by a funnel like opening called ostia. Unlike the male frog, the female frog has separate genital ducts distinct from ureters. Posteriorly the oviducts dilated to form **ovisacs** before they open into cloaca. Ovisacs store the eggs